

# RGB LED

DO

OUTPUT

LIGHT

## What does it do?

An RGB LED puts red, green and blue lights inside one clear package.

Varying the brightness of the three colours mixes millions of possible colours.

A common-cathode LED has one shared negative leg and needs one resistor for each colour.

## Uses

### Status light



**DETECT**  
Read the  
machine state

**DECIDE**  
Choose a  
status colour

**DO**  
Mix the LED  
colour

### Mood crystal



**DETECT**  
Read a  
mysterious  
input

**DECIDE**  
Pick the  
crystal's  
mood

**DO**  
Glow that  
colour



**Clear lens**  
Contains three  
LEDs

**Long leg**  
Common  
cathode

**R**  
One 220 ohm  
resistor each

# Code

## ● Raspberry Pi

```
from gpiozero import RGBLED
light = RGBLED(red=17, green=27, blue=22)
light.color = (1, 0, 1)
```

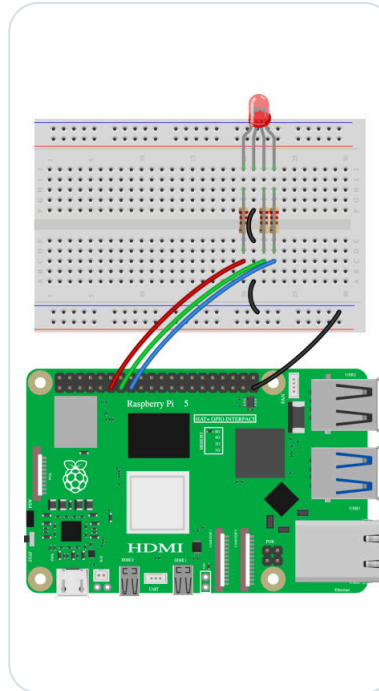
## ● Raspberry Pi Pico

```
from picozero import RGBLED
light = RGBLED(red=15, green=13, blue=10)
light.color = (255, 0, 255)
```

Pi colour values run from 0 to 1; picozero colour values run from 0 to 255. These examples make magenta.

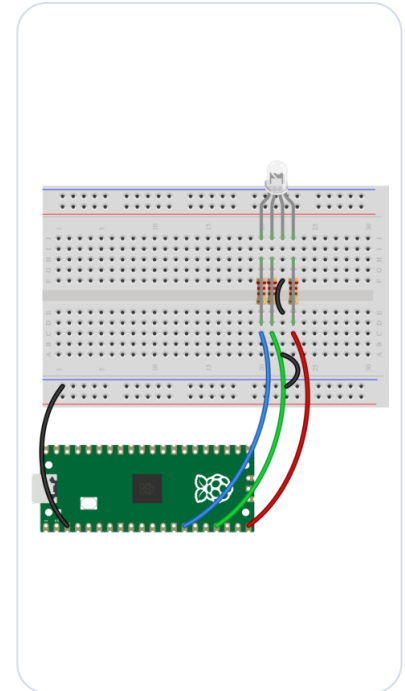
# Wiring

## RASPBERRY PI 5



R → **GPIO17**, G → **GPIO27**, B → **GPIO22**;  
common → **GND**

## PICO



R → **GP15**, G → **GP13**, B → **GP10**;  
common → **GND**